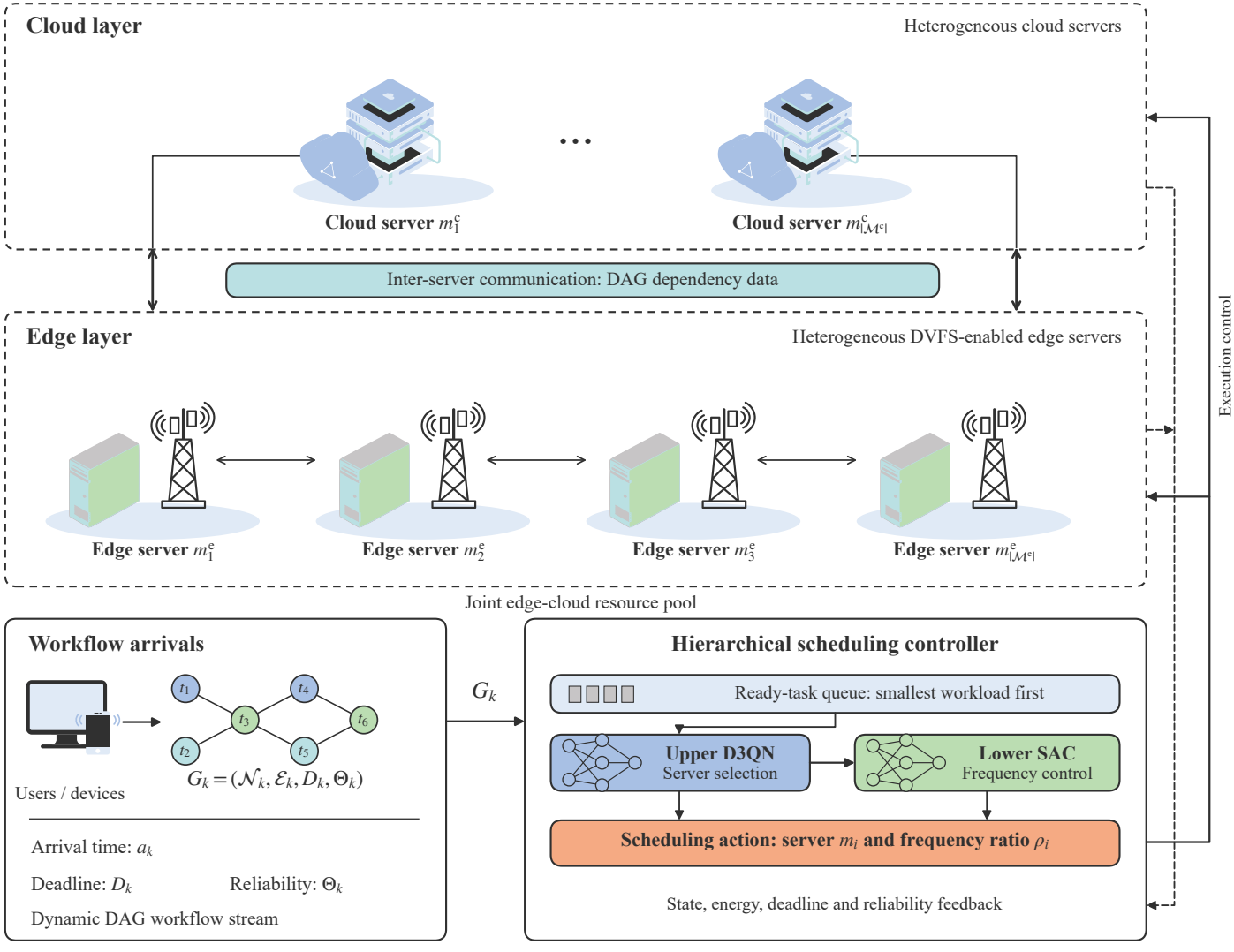

Cloud layer
Heterogeneous cloud servers
Cloud server m_1^c
Cloud server $m_{|\mathcal{M}^c|}^c$
Inter-server communication: DAG dependency data
Edge layer
Heterogeneous DVFS-enabled edge servers
Edge server m_1^e
Edge server m_2^e
Edge server m_3^e
Edge server $m_{|\mathcal{M}^e|}^e$
Execution control
Joint edge-cloud resource pool
Workflow arrivals
t_1 t_2 t_3 t_4 t_5 t_6
$G_k = (\mathcal{N}_k, \mathcal{E}_k, D_k, \Theta_k)$
Users / devices
Arrival time: a_k
Deadline: D_k
Reliability: Θ_k
Dynamic DAG workflow stream
G_k
Hierarchical scheduling controller
Ready-task queue: smallest workload first
Upper D3QN
Server selection
Lower SAC
Frequency control
Scheduling action: server m_i and frequency ratio ρ_i
State, energy, deadline and reliability feedback